INSTITUTO NACIONAL DE EDUCACIÓN LOURDES — INDEL

Bachillerato Técnico Vocacional en Desarrollo de Software

Module 3.6 — English Conversation on IT Systems Maintenance

ANNEX A
SYSTEM ANATOMY MAP

Individual Activity 02 — Windows Preventive Maintenance · Day 1

Anexo A — Mapa de anatomía del sistema · Actividad Individual 02 · Día 1

Student / Estudiante	Estela Guadalupe Amaya Mata
Grade and section / Grado y sección	3.º Desarrollo de Software "A"
Computer ID / Equipo	LAPTOP-7SKAH6B6
Location / Ubicación	INDEL
Owner or user / Propietario	MINEDUCYT
Date / Fecha	2026-08-18
Teacher / Docente	Ing. Javier Arturo García Mineros

1. System identification

1. Identificación del sistema

ITEM	VALUE FOUND ON THE MACHINE
Windows edition	Windows 11 Pro Education
Version / Build	226631.5039
Installation date	17/02/2023
Device name	LAPTOP-7SKAH6B6
Processor	11th Gen Inter(R) Core (TM)

Installed RAM	4.00 GB
System type	x64-based PC
BIOS mode / Secure Boot	UEFI/ Off
Drive C: total and free space	118 GB total, 3.25 GB free
File system of C:	NTFS
System up time	0 days 4 hours
Support status of this Windows version	Supported

2. Component map / Mapa de componentes

COMPONENT	FUNCTION — IN ENGLISH	FUNCIÓN — EN ESPAÑOL	TOOL USED TO INSPECT IT
Kernel	Core of the OS that manages system resources and hardware communication.	Núcleo del sistema operativo que gestiona los recursos y el hardware.	Msinfo32
Device drivers	Software that allows the operating system to communicate with hardware devices.	Software que permite al sistema operativo comunicarse con el hardware.	Device Manager
Services	Background processes that run without user interaction to perform core system tasks.	Procesos en segundo plano que ejecutan tareas del sistema sin intervención del usuario	Services.msc
Processes	Programs currently executing in memory with their own allocated resources.	Programas en ejecución en la memoria con recursos asignados.	Task Manager

Shell (explorer.exe)	User interface that provides access to the file system, taskbar, and desktop.	Interfaz gráfica que proporciona acceso a archivos, barra de tareas y escritorio.	Task Manager
Registry	Hierarchical database storing configuration settings for Windows and installed applications.	Base de datos jerárquica que guarda configuraciones de Windows y aplicaciones.	regedit
File system (NTFS)	Structure used to store, organize, and manage files on the drive with security permissions.	Estructura para almacenar, organizar y gestionar archivos con permisos de seguridad.	File explorer
Startup apps	Programs configured to launch automatically when Windows boots up.	Programas configurados para iniciarse automáticamente al arrancar Windows.	Task Manager
Windows Update	Service that downloads and installs security patches and operating system features.	Servicio que descarga e instala parches de seguridad y actualizaciones.	Settings
Windows Security	Built-in protention suite against viruses, malware, and network threats.	Suite de protección integrada contra virus malware y amenazas de red.	Windows Security
Event logs	Records of systems, application, and security events used for troubleshooting.	Registros de eventos del sistema y aplicaciones para diagnóstico de errores.	Eventvwr.msc
Virtual memory (pagefile)	Space on the hard drive used as RAM extension when physical memory is full.	Espacio del disco usado somo extensión de RAM cuando la memoria física se llena.	Sysdm.cpl

3. Boot sequence / Secuencia de arranque

#	STAGE	WHAT HAPPENS — IN YOUR OWN WORDS (ENGLISH)
1	Power on / POST	Hardware self-test initializes power, checking CPU, RAM, and essential components.

2	UEFI / BIOS	Initializes system hardware and locates the boot device to start the operating system.
3	Windows Boot Manager	Reads boot configuration data (BCD) and selects the operating system partition.
4	OS Loader	Load essential drivers into memory and prepares the system to initialize the kernel.
5	Kernel initialization	Loads core Windows services, hardware drivers, and system files into RAM.
6	Session Manager / Logon	Stars user subsystem serves and presents the Windows login screen.
7	Shell and startup apps	Loads explorer.exe interface, taskbar, desktop background, and startup programs.

4. Laboratory observations / Observaciones de laboratorio

#	WHAT I OBSERVED	RESULT / VALUE	IS THIS NORMAL? WHY? (ENGLISH)
1	Edition, version and build	Windows 11 Pro Education 23H2/22631.5039	Yes. It fully updated and supported for school.
2	System up time and RAM in use	Up time: RAM:3.67 GB used (4 GB total)	Yes. System processes require substantial memory resources during operation.
3	Top 3 processes by memory	Chrome, Windows Explorer, Antivirus	Yes. Web browsers and system tools consume high RAM on 4 setups.
4	Startup apps with high impact	OneDrive, Microsoft Teams. Security Health	Yes. Cloud services and security apps automatically launch at startup.
5	Windows Update service status	Running (Automatic)	Yes. Maintains systems security patches and background system maintenance.

6	Devices with a warning icon	None / No warning	Yes. All hardware drivers are working correctly without driver errors.
7	Partitions and file system of C:	C: Drive / NTFS / 118 GB total	Yes. Standard file system format used for primary Windows partitions.
8	Errors in the System log (7 days)	Low number of minor non-critical warnings	Yes. Occasional warnings occur during routine system startup procedures.
9	Last antivirus scan and status	No actions required	Yes. Windows Defender real-tin.
10	Pending Windows updates	System is up to date	Yes. Latest system updates and patches have been applied successfully.

5. Component descriptions / Descripción de componentes

5.1 English version

1. The kernel acts as the core of the operating system managing hardware resources and system memory.

2. Device drivers allow the operating system to communicate properly with physical hardware components.

3. Windows services execute key system funtions in the background without requiring user interaction.

4. The shell interface creates the desktop environment and taskbar to interact with stored files.

5. NTTFS file system organizes drive data securely while maintaining file permissions and access controls.

5.2 Versión en español

1. El núcleo actúa como la parte central del sistema operativo administrando el hardware y la memoria.

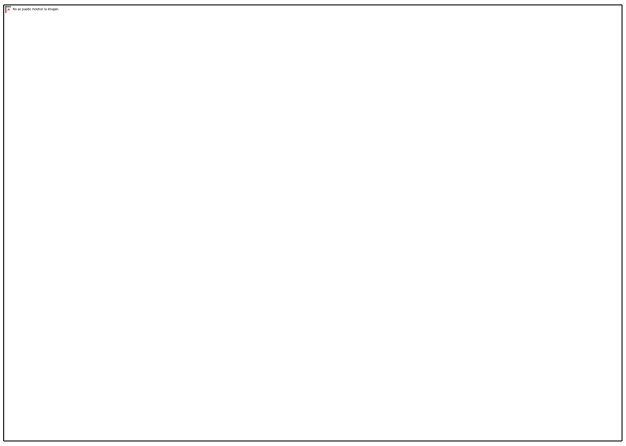
2. Los controladores permiten que el sistema operativo se comunique correctamente con el hardware físico.

3. Los servicios de Windows ejecutan funciones claves en segundo plano sin requerir intervención del usuario.

4. La interfaz del Shell crea el entorno del escritorio y la barra de tareas para interactuar con los archivos.

5. El sistema de archivos NTFS organiza los datos del disco de forma segura manteniendo los permisos.

6. Evidence index / Índice de evidencias

EVIDENCE	CAPTION IN ENGLISH	TOOL / SCREEN
EV-01	System properties showing Windows 11 specs, CPU, and RAM allocation	
EV-02	Local disk C: showing available total storage space and current capacity.	
EV-03	Task Manager performance view showing active processes and system memory usage.	

EV-04	Windows Security overview displaying active protection status and scan options.	
EV-05	Device Manager tree listing hardware devices and driver operational status.	

Instruction: paste each numbered screenshot below this table, in order, keeping its English caption underneath.
Pega cada captura numerada debajo de esta tabla, en orden, conservando su pie de foto en inglés.

7. Statement of authorship / Declaración de autoría

I declare that I personally carried out this inspection on the computer described above and that the data and screenshots presented are real.

Declaro que realicé personalmente esta inspección en el equipo descrito y que los datos y capturas presentados son reales.

_____	_____
Estela Guadalupe Amaya Mata	Ing. Javier A. García Mineros
Student signature / Firma del estudiante	Teacher / Docente